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Module - 4

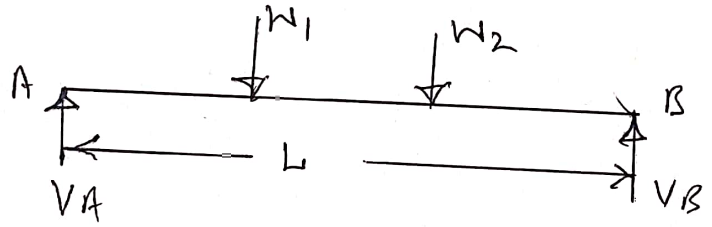
Support Reactions:-

Types of loads:-

i) Concentrated or point load
ii) Uniformly distributed load.
iii) Uniformly varying load.

i) Concentrated or point load:

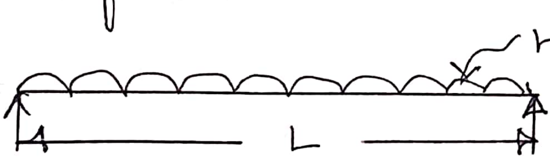
A load acting at a point on a beam is known as concentrated or point load.



In the above figure W_1 and W_2 are point loads.

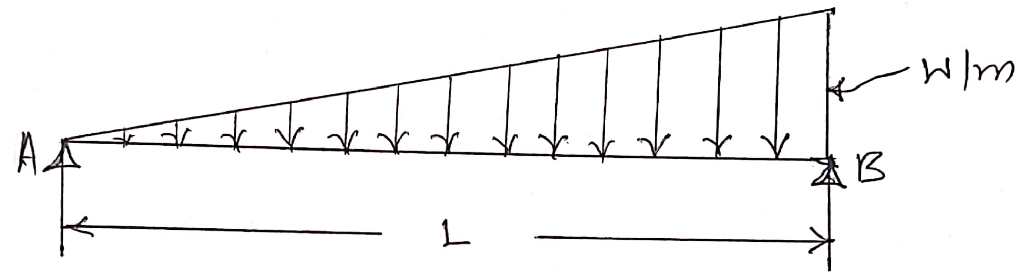
ii) Uniformly distributed load:-

If a beam is loaded in such a way that each unit length of the beam carries same intensity of load then that type of load is known as uniformly distributed load.



iii> Uniformly varying load:

If a beam is loaded in such a way that each unit length of the beam carries uniformly varying intensity of loading, then this type of loading is known as uniformly varying load.



Types of Supports:-

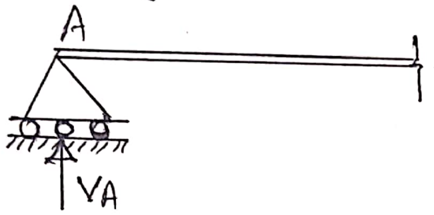
i> Simple support
ii> Roller support
iii> Hinge Support
iv> Fixed Support.

i> Simple support:- If a beam rests simply on the support, then support is called simple support.

In the above figure VA & VB are simple supports

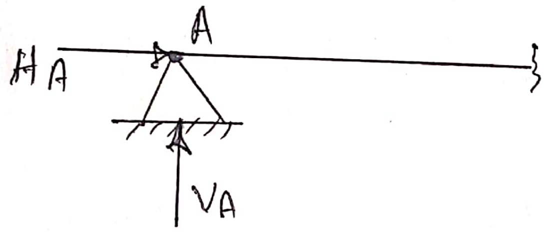
ii) Roller support:-

If a beam is supported on rollers then such support is called as roller support.



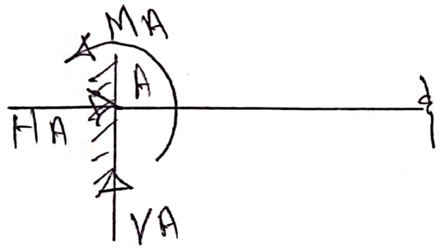
iii) Hinge support:-

If a beam is supported on a hinge or pin then such supports are called as hinged or pinned support. The hinge support develops two reactions.



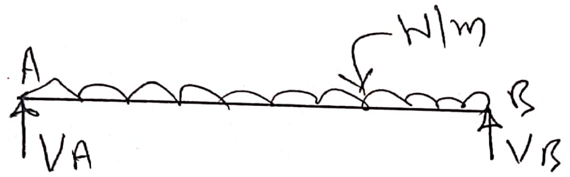
iv) Fixed support:-

If the end of the beam is fixed or built in, then such support is called as fixed support. The fixed support develops three reactions.



Support Reaction:-

When a number of forces acting on a beam then the supports of the beam develops the reactions are called support reactions.

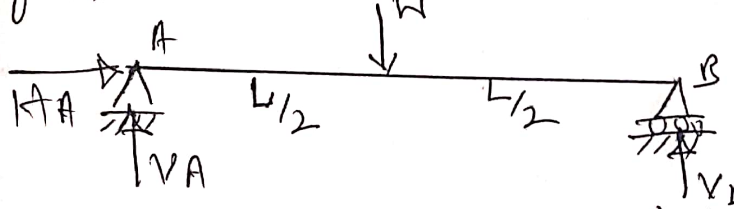


In the above figure VA and VB are called support reactions.

Determinate and indeterminate beams:-

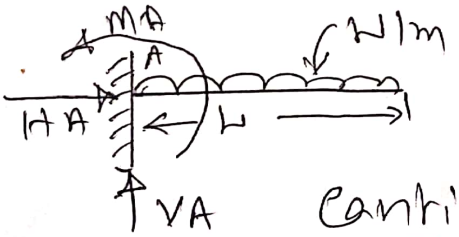
In a beam if it is possible to determine the support reactions using equilibrium equations, ie $\Sigma V = 0$, $\Sigma H = 0$ and $\Sigma M = 0$, then the beam is considered as determinate beam.

Eg: 1.



Simply Supported Beam

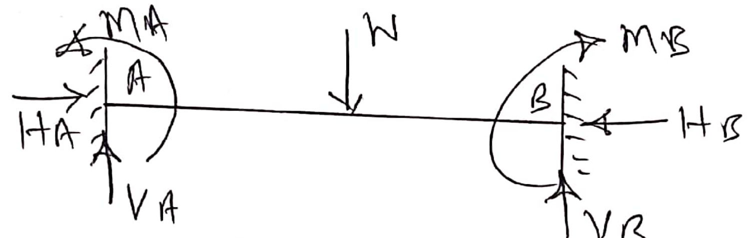
Eg: 2



Cantilever beam

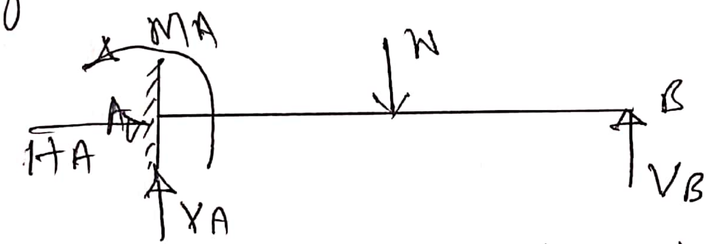
In a beam if it is not possible to determine the support reactions using equilibrium equations ie $\Sigma V=0$, $\Sigma H=0$ and $\Sigma M=0$, then the beam is considered as indeterminate beam.

Eg: 1



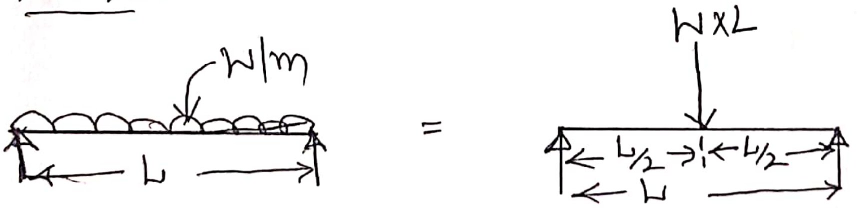
Fixed beam

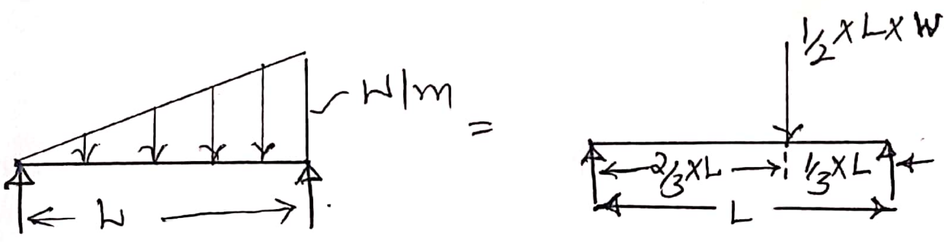
Eg: 2



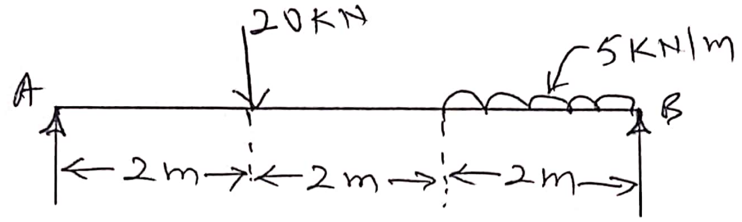
Proped Cantilever beam

Note:-

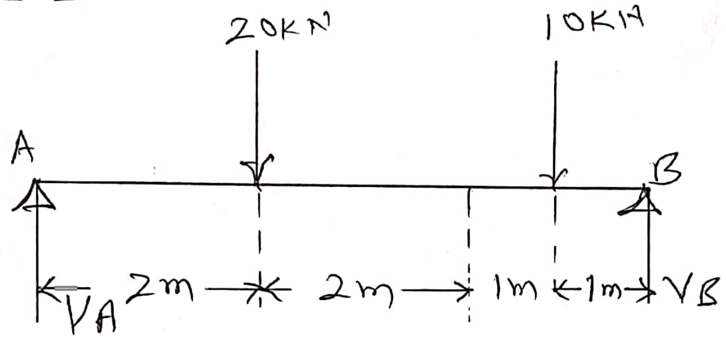




1) Determine reactions at the support for the beam shown below.



Solution:-



$\Sigma MA = 0$

$20 \times 2 + 10 \times 5 - VB \times 6 = 0$

$VB = 15 KN$

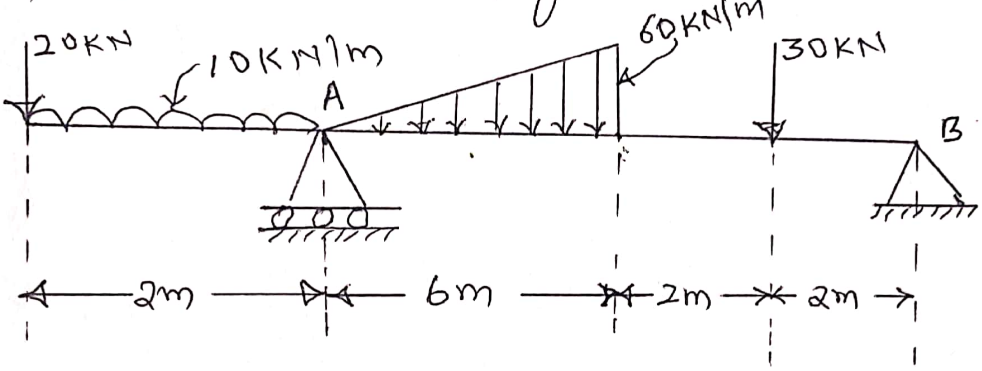
$\Sigma V = 0$

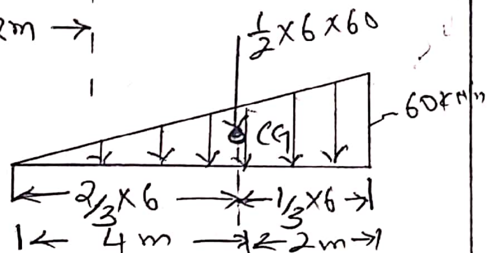
$VA - 20 - 10 + VB = 0$

$VA - 20 - 10 + 15 = 0$

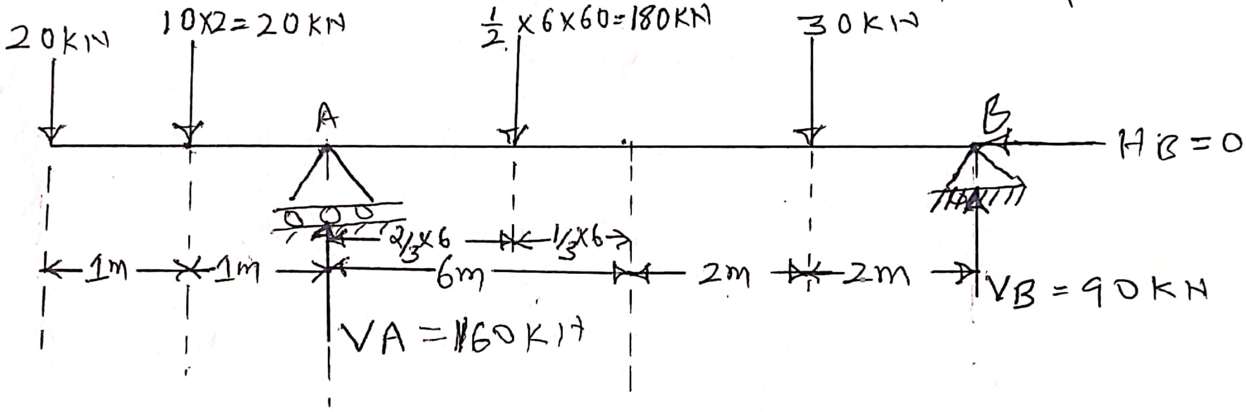
$VA = \underline{\underline{15}} KN$

1) Determine reactions at supports for the beam shown in fig below.





Solution:-



$\Sigma M_A = 0$ (↻+ ↺−)

$-20\times2 - 20\times1 + 180\times4 + 30\times8 - V_B\times10 = 0$

$V_B = 90\,KN$

$\Sigma V = 0$ (↑+ ↓−)

$-20 - 20 + V_A - 180 - 30 + V_B = 0$

$-20 - 20 + V_A - 180 - 30 + 90 = 0$

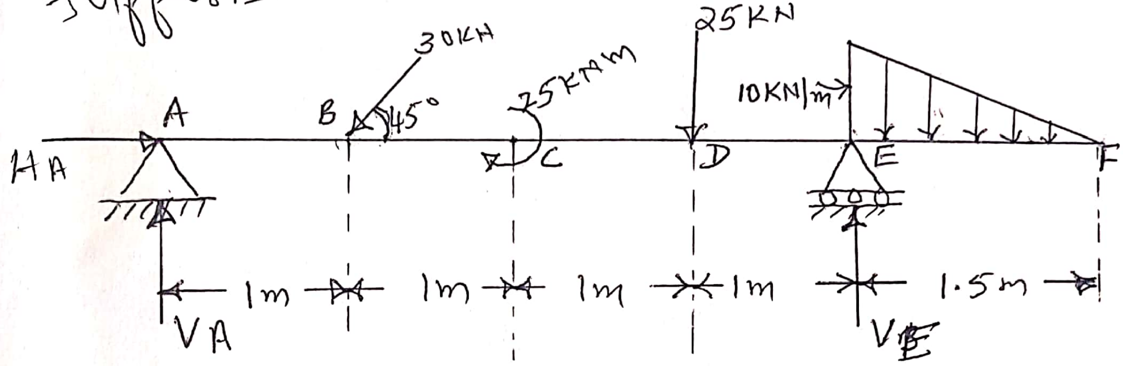
$V_A = 160\,KN$

$\Sigma H = 0$, (→+ ←−)

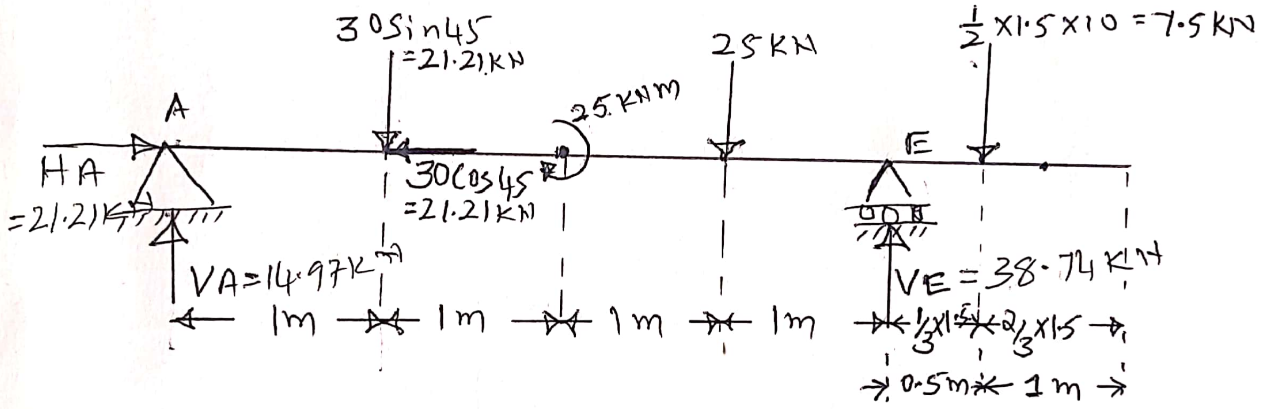
$-H_B = 0$

$\therefore H_B = \underline{0}$

3) A beam ABCDEF is hinged at 'A' and supported on roller at 'E' and carries loads as shown. Determine reactions at supports.



Solution:



$\Sigma M_A = 0$

$21.21 \times 1 + 25 + 25 \times 3 - V_E \times 4 + 7.5 \times 4.5 = 0$

$V_E = 38.74\,KN$

$\Sigma V = 0$ $\quad \uparrow + \downarrow -$

$V_A - 21.21 - 25 + V_E - 7.5 = 0$

$V_A - 21.21 - 25 + 38.74 - 7.5 = 0$

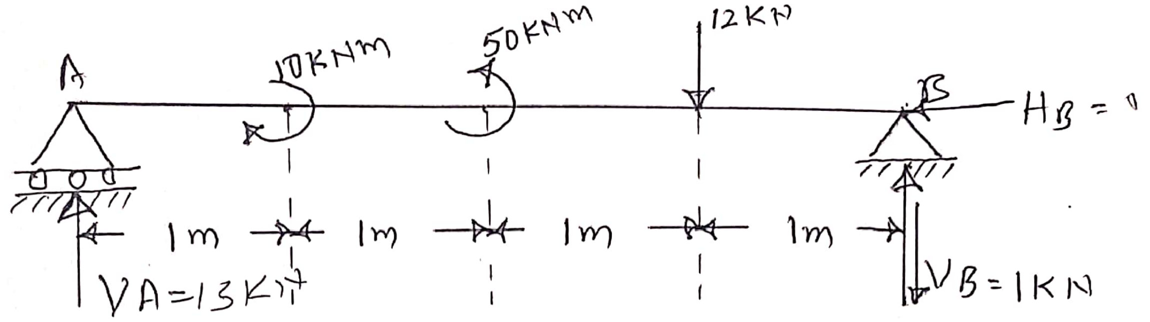
$V_A = 14.97\,KN.$

$\Sigma H = 0$ $\quad \xrightarrow{+} \quad \xleftarrow{-}$

$H_A - 21.21 = 0$

$H_A = 21.21\,KN$

4) Determine the reaction at the support for the beam shown in fig below.



$\Sigma M_A = 0$

$10 - 50 + 12 \times 3 - V_B \times 4 = 0$

$V_B = -1\ KN\ (\downarrow)$

$\Sigma V = 0$

$V_A - 12 - 1 = 0$

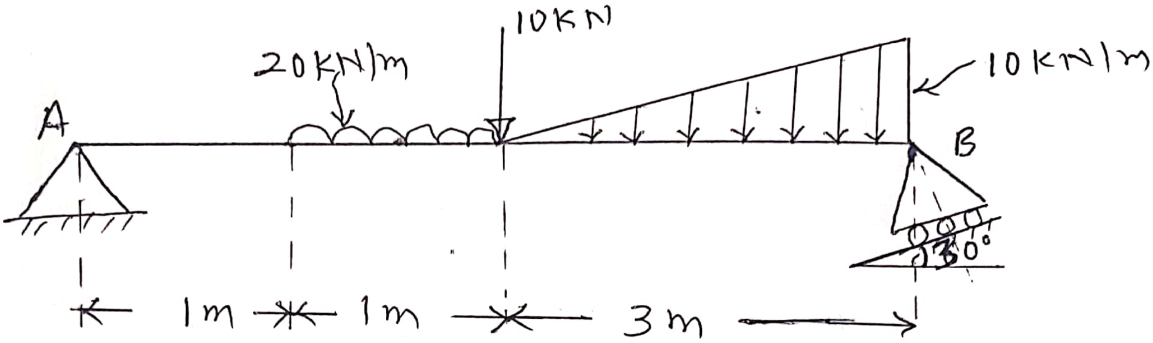
$V_A = 13 KN$

$\Sigma H = 0$

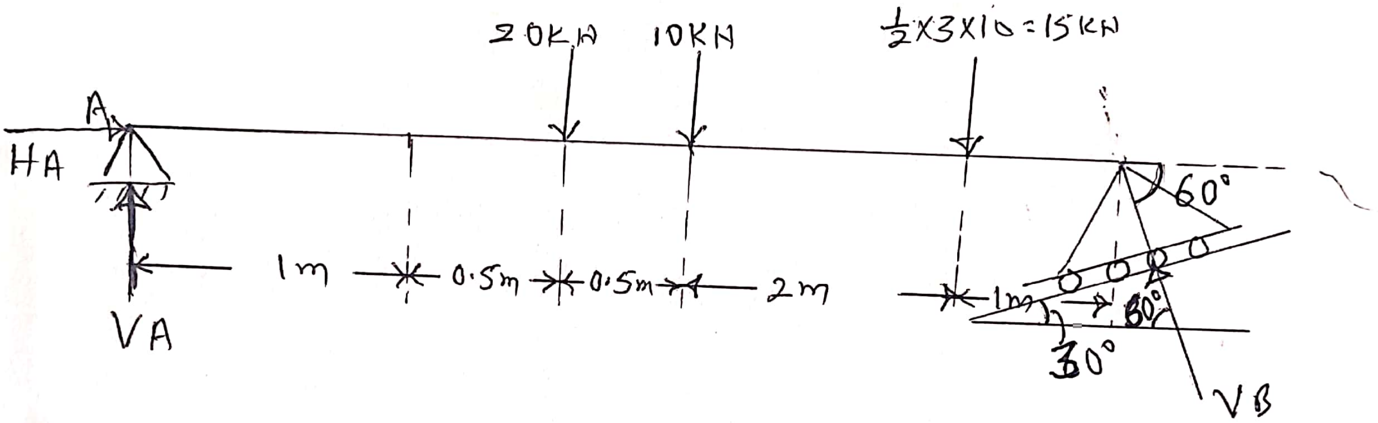
$-H_B = 0$

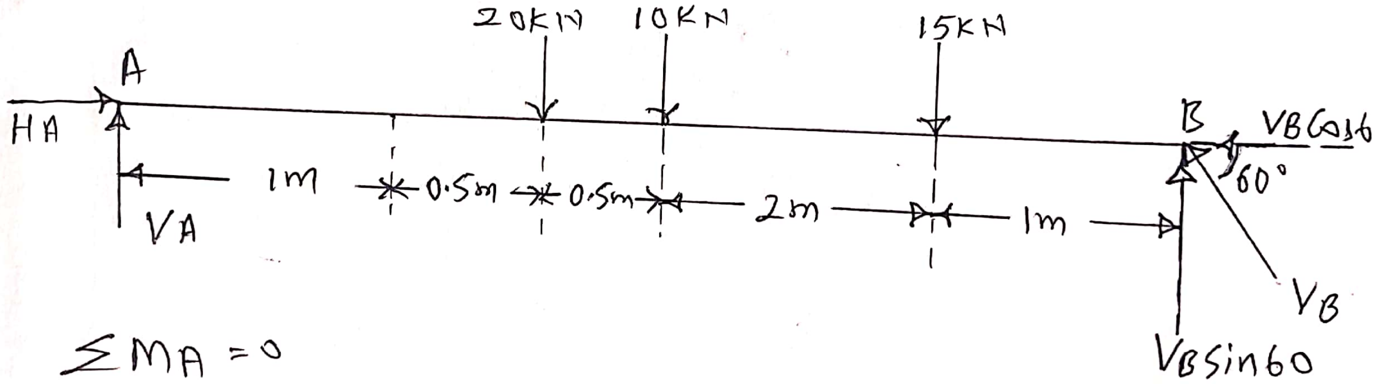
$H_B = 0$

5) Determine support reactions for the beam shown in fig.



Solution:





$\Sigma M_A = 0$

$$20 \times 1.5 + 10 \times 2 + 15 \times 4 - V_B \sin 60 \times 5 = 0$$

$$V_B = 25.4 \text{ KN}$$

$\Sigma V = 0$

$$V_A - 20 - 10 - 15 + V_B \sin 60 = 0$$

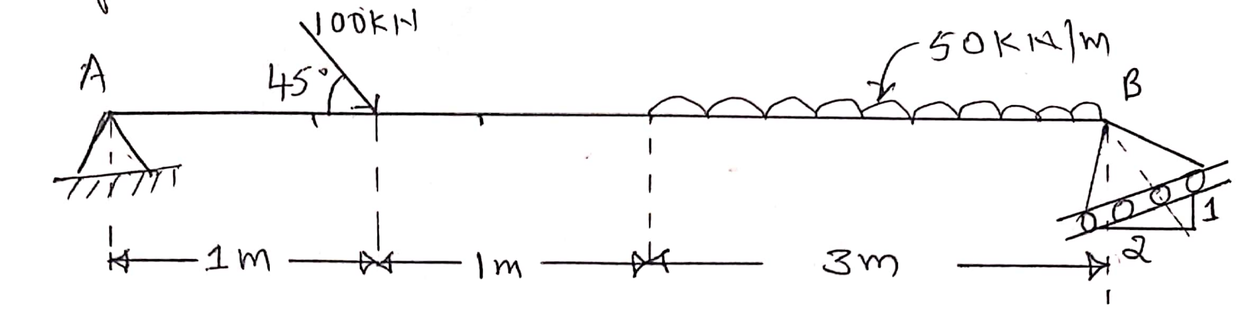
$$V_A = 23 \text{ KN}.$$

$\Sigma H = 0$

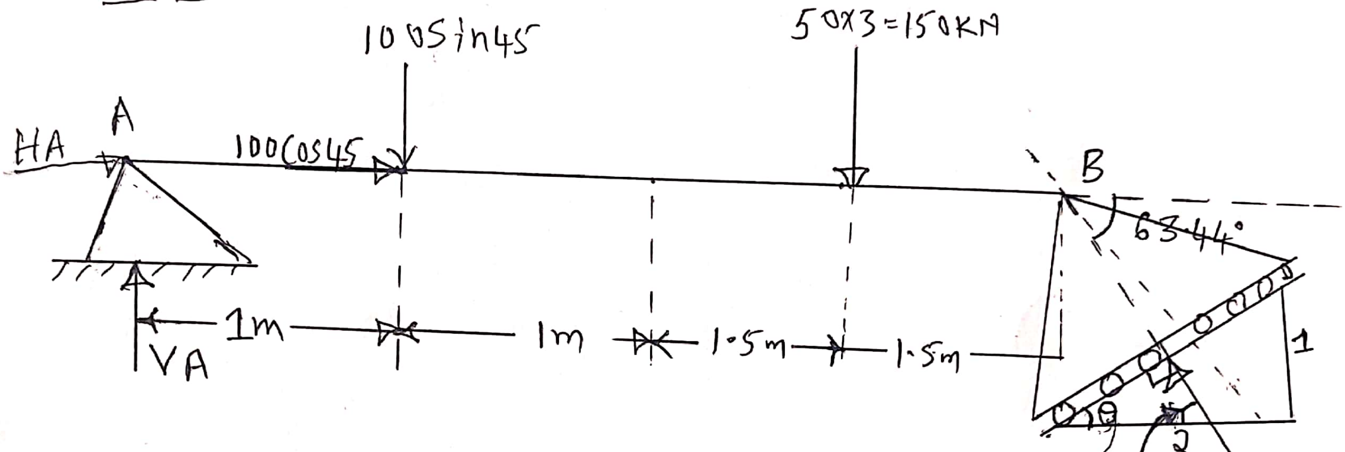
$$H_A - V_B \cos 60 = 0$$

$$H_A = 12.7 \text{ KN}$$

6) Determine the reactions at supports for the beam shown below.

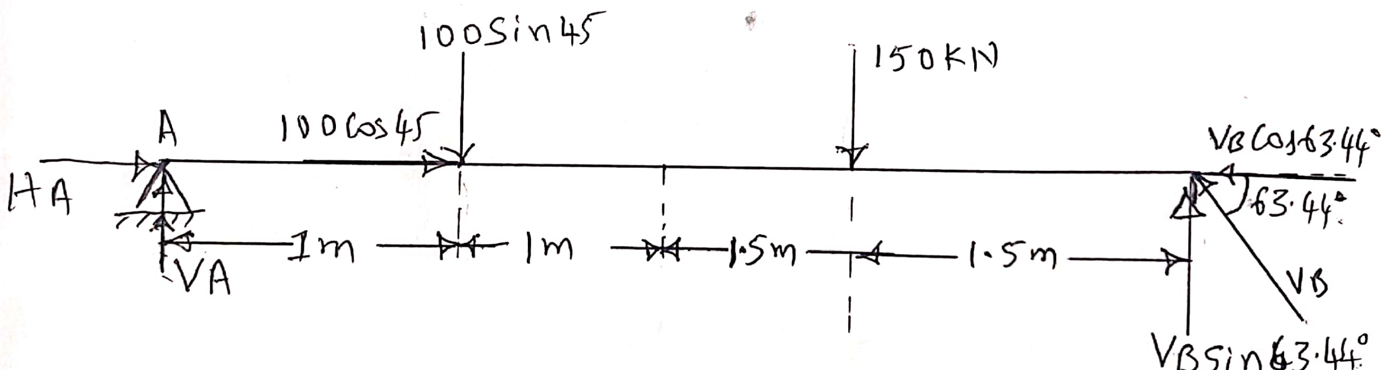


Solution:-



At support B, $\tan\theta = \frac{1}{2}$

$\theta = \cancel{63.44} \ 26.56°$



$\Sigma M_A = 0$

$100 \sin 45 \times 1 + 150 \times 3.5 - V_B \sin 63.44 \times 5 = 0$

$V_B = \underline{133.26 \text{ KN}}$

$\Sigma V = 0$

$V_A - 100 \sin 45 - 150 + 133.26 \sin 63.44 = 0$

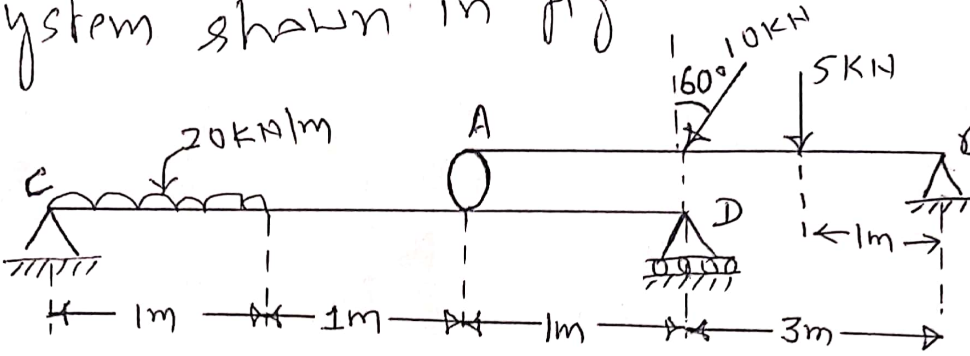
$V_A = 101.51 \text{ KN}$

$\Sigma H = 0$

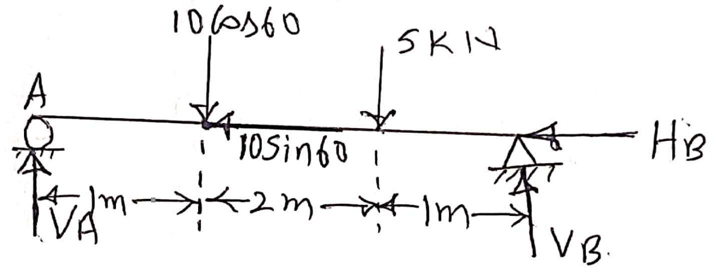
$H_A + 100 \cos 45 - 133.26 \cos 63.44 = 0$

$H_A = \underline{-11.12} \text{ KN } (\leftarrow)$

7) Determin the reactions at supports for the system shown in fig.



Consider upper beam



$\Sigma M_A = 0$

$10\cos 60 \times 1 + 5 \times 3 - V_B \times 4 = 0$

$V_B = 5KN$

$\Sigma V = 0$

$V_A - 10\cos 60 - 5 + V_B = 0$

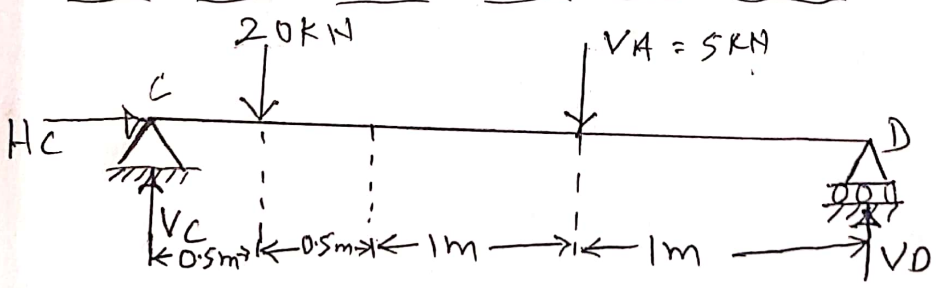
$V_A = 5KN$

$\Sigma H = 0$

$-H_B - 10\sin 60 = 0$

$H_B = -8.66KN \ (\rightarrow)$

Now consider lower beam



$\Sigma M_C = 0$

$20 \times 0.5 + 5 \times 2 - V_D \times 3 = 0$

$V_D = 6.67KN$

$\Sigma V = 0$

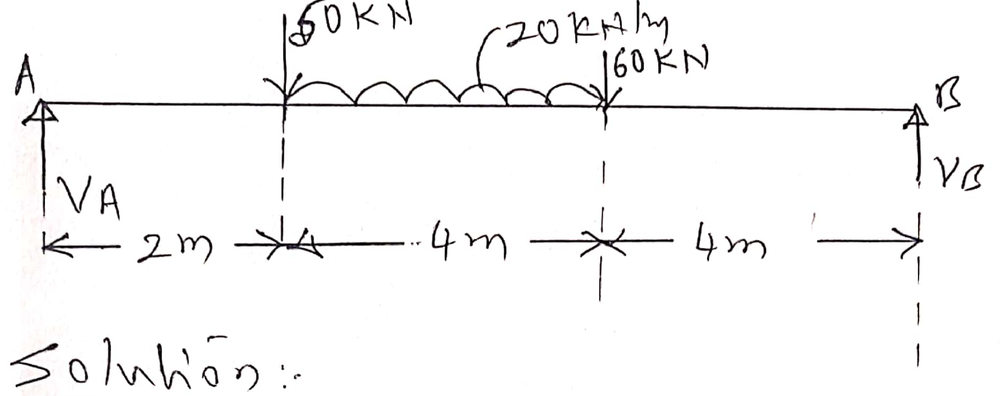
$V_C - 20 - 5 + 6.67 = 0$

$V_C = 18.33KN$

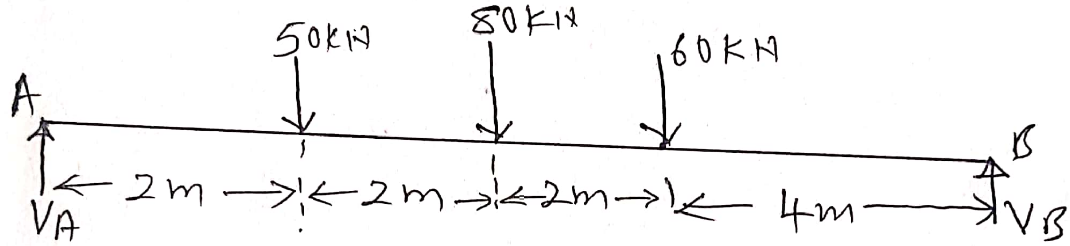
$\Sigma H = 0$

$H_C = 0$

8) Determine the support reactions for the beam shown in fig.



Solution :-



$\Sigma MA = 0$

$50 \times 2 + 80 \times 4 + 60 \times 6 - VB \times 10 = 0$

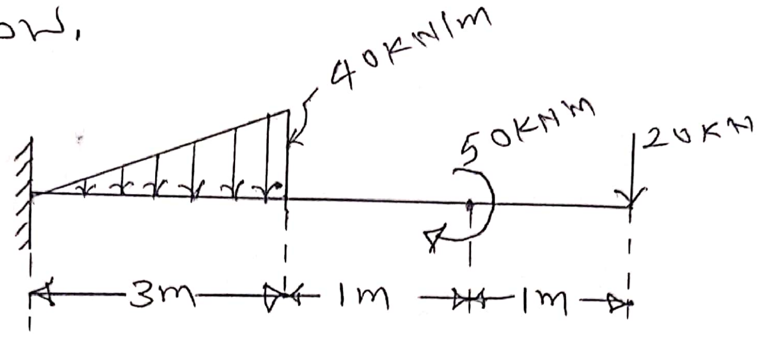
$VB = 78 KN$

$\Sigma V = 0$

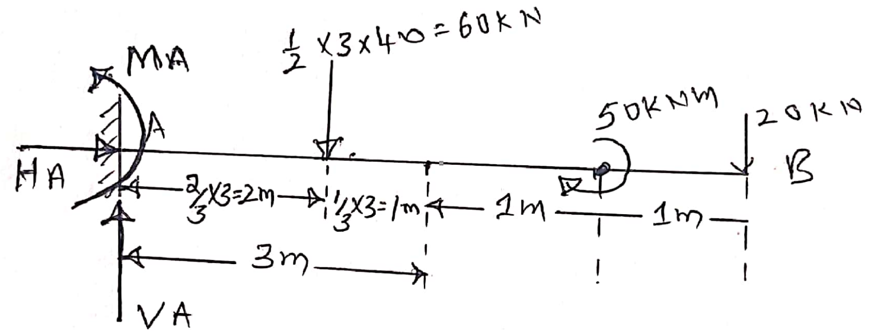
$VA - 50 - 80 - 60 + 78 = 0$

$VA = 112 KN$

9) Determine reaction components for the cantilever beam shown in fig below.



Solution:-



[In cantilever beam one end is fixed and other end is free i.e end B is free and no support at 'B'.
At fixed end there are three reactions i.e MA, HA and VA]

$\Sigma MA = 0$

$$-MA + 60\times2 + 50 + 20\times5 = 0$$

$$MA = \underline{270} \; KNm$$

$\Sigma V = 0$

$$VA - 60 - 20 = 0$$

$$VA = 80 \; KN$$

$\Sigma H = 0$

$$HA = \underline{0}$$

Analysis of trusses:-

Assumptions made in the analysis of trusses.

i) The members of the truss are straight.

ii) The cross section of members are uniform.

iii) Forces are acts at joint only.

iv) All members are pin jointed.

v) All members are rigid.

vi) All members are tensile or compressive in nature.

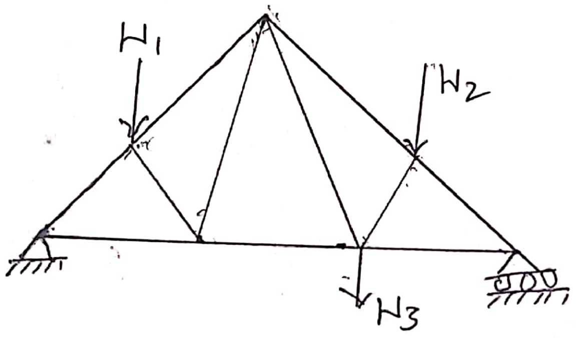
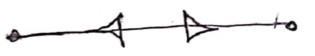


Fig: Truss

 (Tensile force)

 (Compressive force)

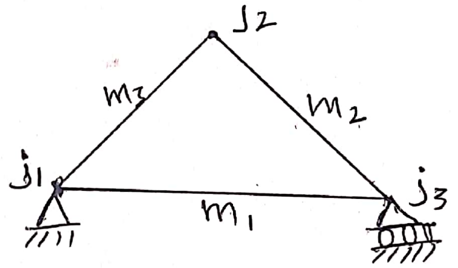
Types of trusses

i) Perfect truss

ii) Deficient truss

iii) Redundent truss.

i) **Perfect truss**:- A pin jonted truss which has got sufficient members to resist load without undergoing appreciable deformation in shape is called as perfect truss.
It is also satisfies $m = 2j - 3$ where m = Number of members, j = Number of joints.



In the above truss

$m = 3, \quad j = 3$

$m = 2j - 3 \quad -①$

$m = 3$

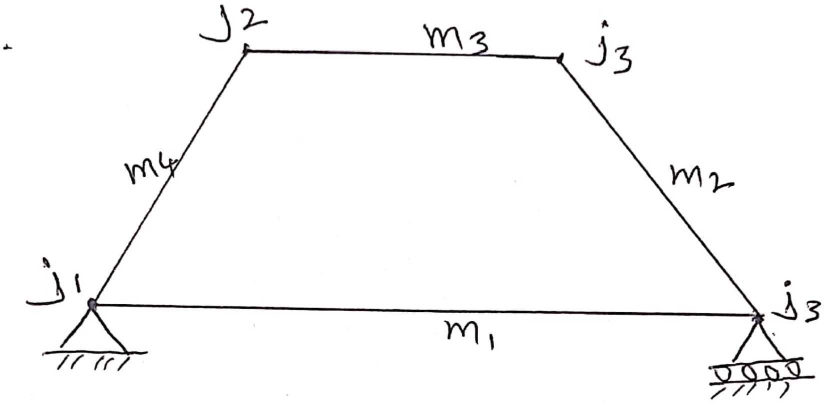
$2j - 3 = 2 \times 3 - 3 = 3$

The eqnation ① is satisfied, hence the truss is perfect truss.

ii) Deficient truss:-

A truss is said to be deficient if the number of members in it are less than that require for a perfect truss.

Here $m < 2j-3$



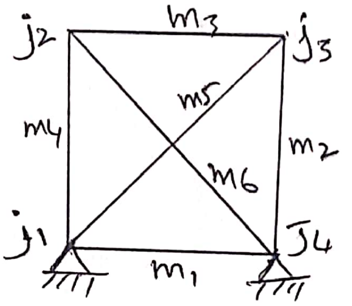
Here $m = 4$
$j = 4$

We have $m = 2j-3$

$2j-3 = 2\times4-3 = 5$

Here $m < 2j-3$, hence the truss is difecient truss.

iii) Redundent truss:- A truss is said to be redundent if the number of members in it are more than that required for a perfect truss. Here $m > 2j-3$.

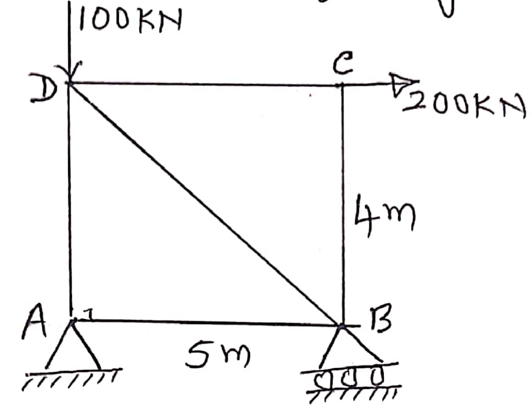


Here $m = 6$
$j = 4$

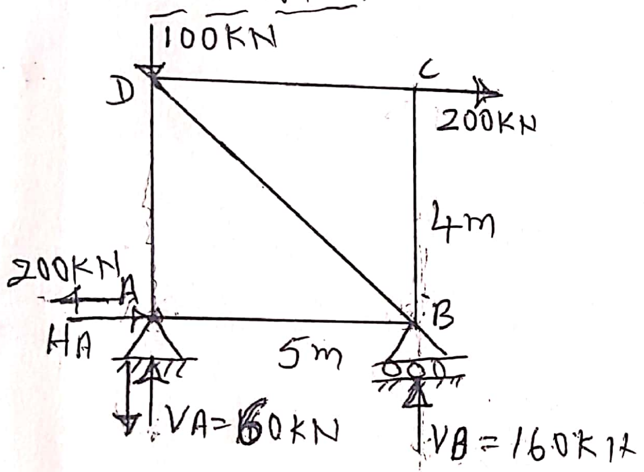
$2j-3 = 2\times4-3 = 5$

$\therefore$ $m > 2j-3$, hence the truss is redundent.

① Find the forces in the members of the truss shown in fig by method of joints.



Solution



$\Sigma M_A = 0$ (↻+ ↺−)

$200 \times 4 - V_B \times 5 = 0$

$V_B = 160KN$

$\Sigma V = 0$ (↑+ ↓−)

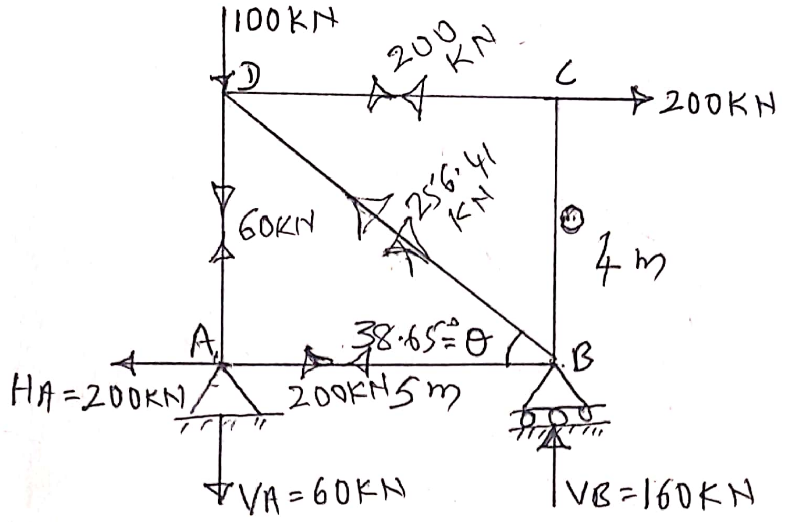
$V_A - 100 + V_B = 0$

$V_A = -60KN$ (↓)

$\Sigma H = 0$ (→+ ←−)

$H_A + 200 = 0$

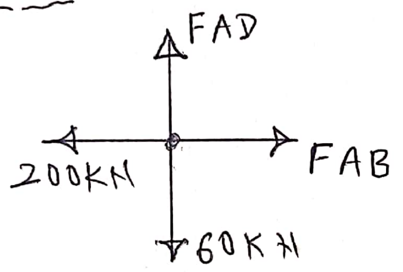
$H_A = -200KN$ (←)



$\tan\theta = \frac{4}{5}$

$\theta = 38.65°$

Joint-A



$\Sigma V = 0 \quad (\uparrow + \downarrow -)$

$F_{AD} - 60 = 0$

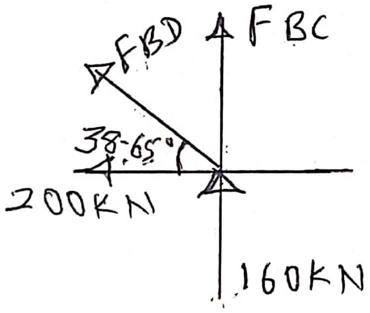
$F_{AD} = 60\,KN\ (T)$

$\Sigma H = 0$

$F_{AB} - 200 = 0$

$F_{AB} = 200\,KN\ (T)$

Joint - B



$\Sigma H = 0$

$-200 - F_{BD}\cos 38.65 = 0$

$F_{BD} = \underline{-256.41}\,KN\ (C)$

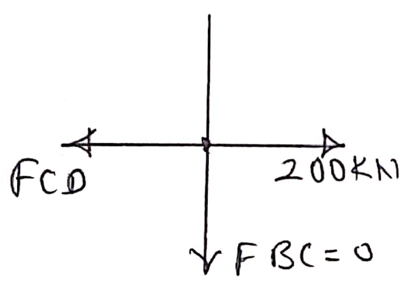
$\Sigma V = 0$

$F_{BD}\sin 38.65 + F_{BC} + 160 = 0$

$-256.41\sin 38.65 + F_{BC} + 160 = 0$

$F_{BC} = 0$

Joint-C :-

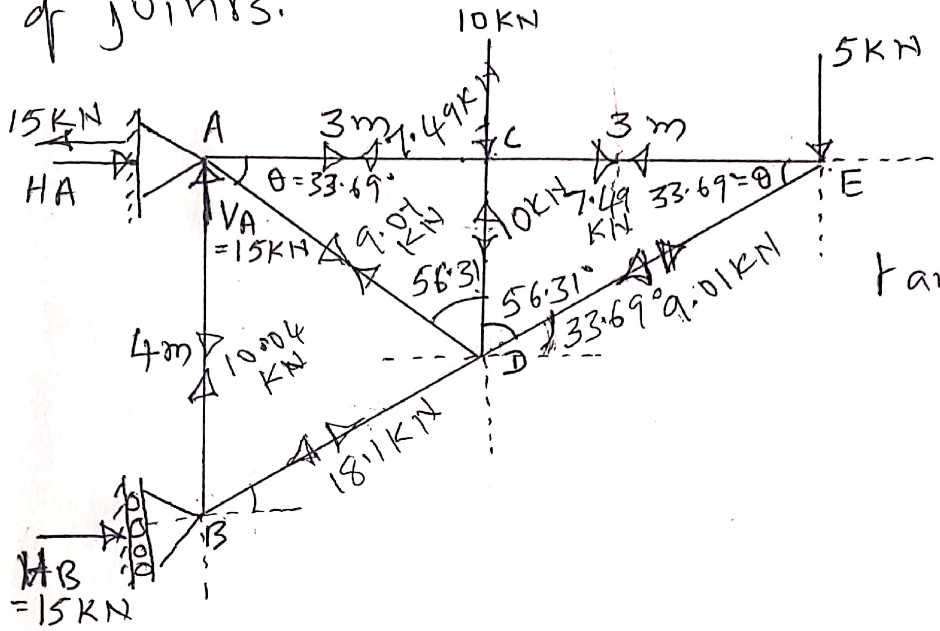


$\Sigma H = 0$

$-F_{CD} + 200 = 0$

$F_{CD} = \underline{200}\,KN\ (T)$

2) Analyse the truss shown in fig by method of joints.



$\tan\theta = \frac{4}{6}$

$\theta = 33.69°$

$\Sigma M_A = 0$

$10 \times 3 + 5 \times 6 - H_B \times 4 = 0$

$H_B = 15\,KN$

$\Sigma H = 0$

$H_A + 15 = 0$

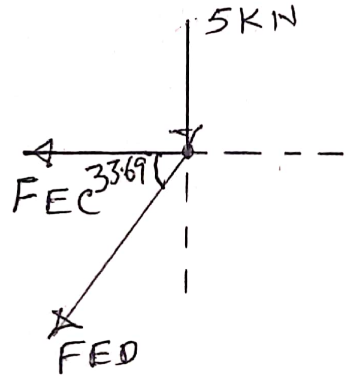
$H_A = -15\,KN\ (\leftarrow)$

$\Sigma V = 0$

$V_A - 10 - 5 = 0$

$V_A = 15\,KN$

Joint-E:-



$\Sigma V = 0 \quad \uparrow + \downarrow -$

$-5 - F_{ED}\sin 33.69 = 0$

$F_{ED} = -9.01\,KN\ (C)$

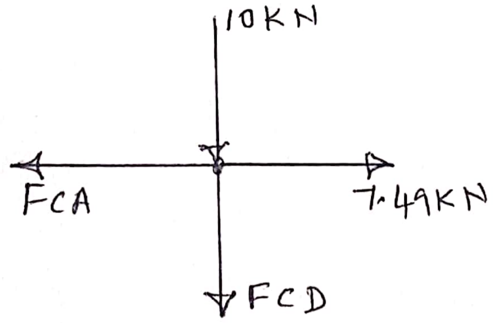
$\Sigma H = 0$

$-F_{EC} - F_{ED}\cos 33.69 = 0$

$-F_{EC} - (-9.01)\cos 33.69 = 0$

$F_{EC} = 7.49\,KN\ (T)$

Joint-C



$\Sigma V = 0$

$-10 - FCD = 0$

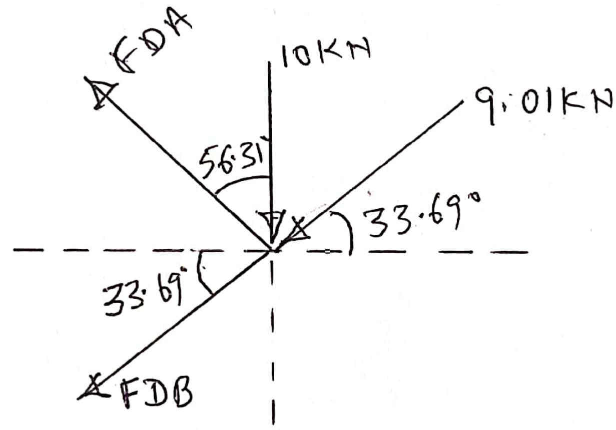
$FCD = -10KN\ (C)$

$\Sigma H = 0$

$-FCA + 7.49\ KN = 0.$

$FCA = 7.49KN\ (T)$

Joint D



$\Sigma V = 0$

$-10 + FDA \cos 56.31 - 9.01 \sin 33.69 - FDB \sin 33.69 = 0$

$-10 + 0.55\ FDA - 4.95 - 0.55\ FDB = 0$

$0.55\ FDA - 0.55\ FDB = 14.95 \quad -①$

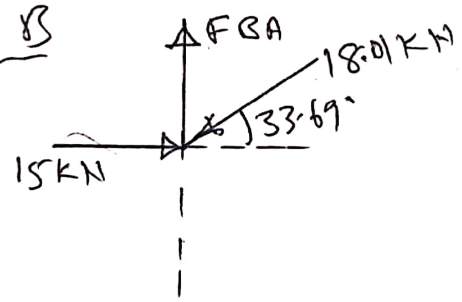
$\Sigma H = 0$

$-FDA \sin 56.31 - 9.01 \cos 33.69 - FDB \cos 33.69 = 0$

$-0.83\ FDA - 0.83\ FDB = 7.49 \quad -②$

Solving ① and ② $\boxed{\begin{array}{l} FDA = 9.07KN\ (T) \\ FDB = -18.10KN\ (C) \end{array}}$

Joint B

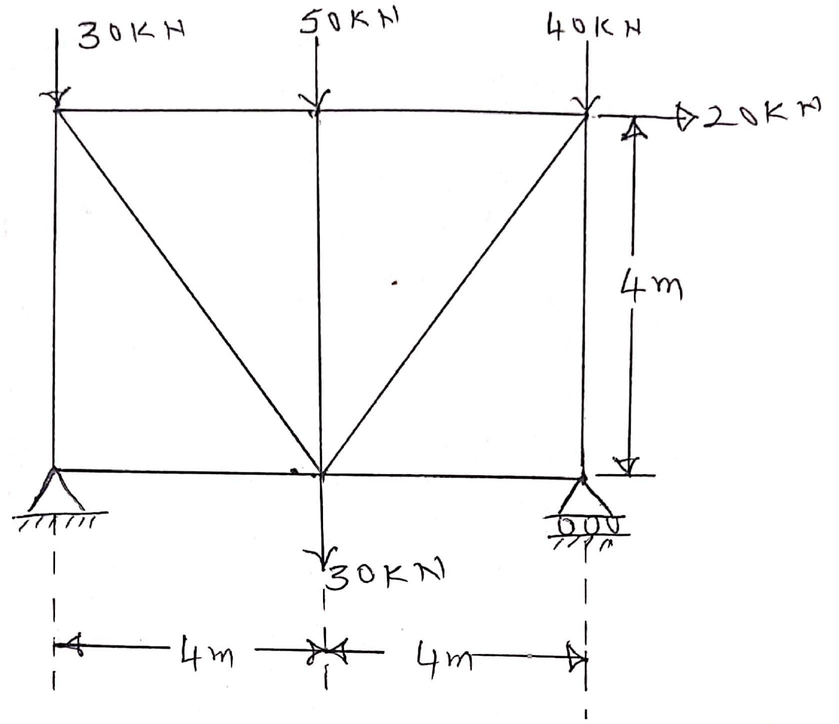


$\Sigma V = 0$

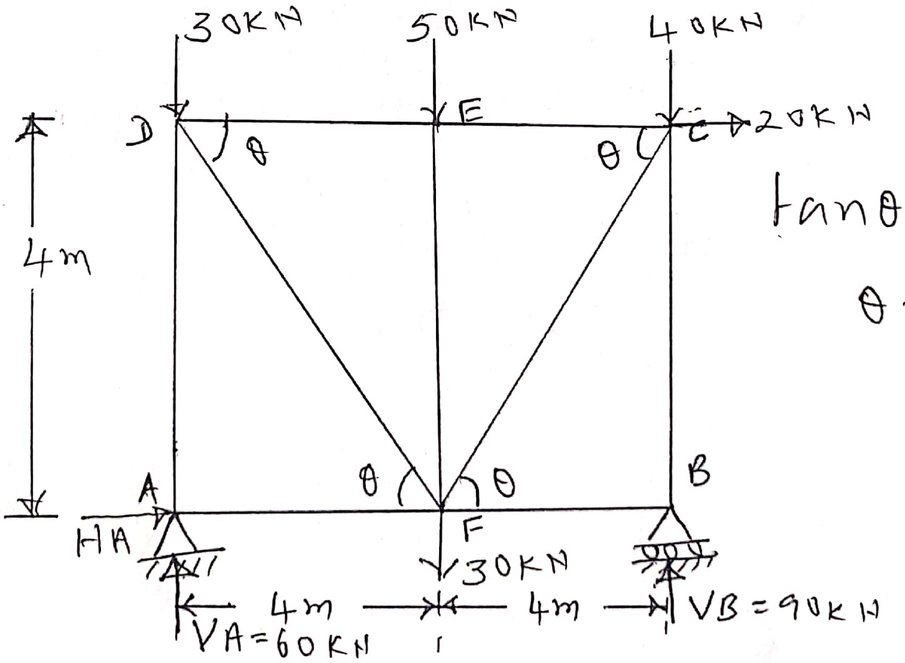
$FBA - 18.1 \sin 33.69 = 0$

$FBA = 10.04KN\ (T)$

3) Analyse the truss shown in fig below by method of joints. Tabulate the member forces.



Solution:-



$\tan\theta = \frac{4}{4}$

$\theta = 45°$

$\Sigma M_A = 0$

$50\times4 + 40\times8 - V_B\times8 + 30\times4 + 20\times4 = 0$

$V_B = 90\,KN$

$\Sigma V = 0$

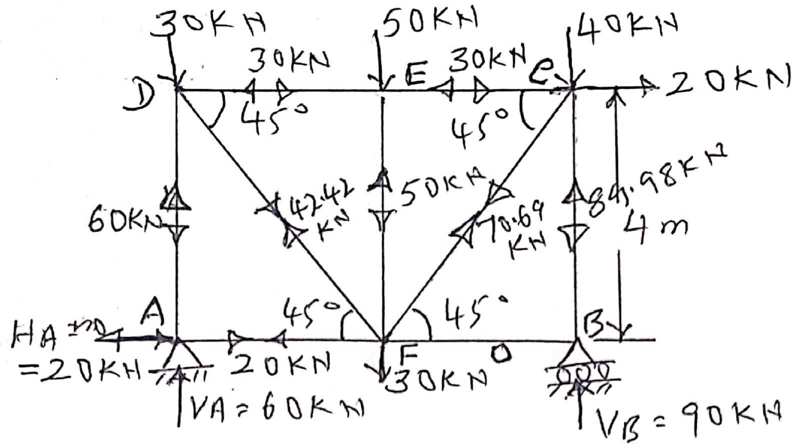
$V_A - 30 - 50 - 40 - 30 - V_B = 0$

$V_A = 60\,KN$

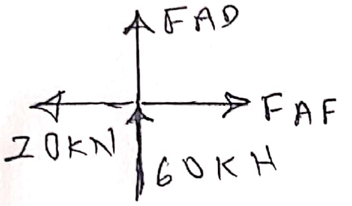
$\Sigma H = 0$

$H_A + 20 = 0$

$H_A = -20\,KN\ (\leftarrow)$



Joint A :-



$\Sigma V = 0$

$60 + FAD = 0$

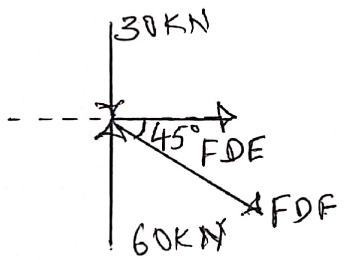
$FAD = -60KN$ (C)

$\Sigma H = 0$

$-20 + FAF = 0$

$FAF = 20KN$ (T)

Joint - D



$\Sigma V = 0$

$-30 + 60 - FDF \sin 45 = 0$

$FDF = 42.42KN$ (T)

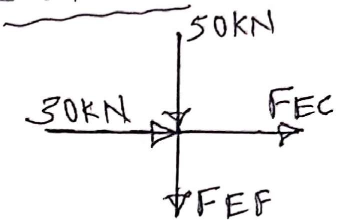
$\Sigma H = 0$

$FDE + FDF \cos 45 = 0$

$FDE + 42.42 \cos 45 = 0$

$FDE = -30KN$ (C)

Joint E :-



$\Sigma V = 0$

$-50 - FEF = 0$

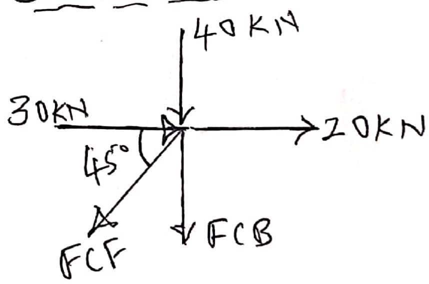
$FEF = -50KN$ (C)

$\Sigma H = 0$

$30 + FEC = 0$

$FEC = -30KN$ (C)

Joint - C :-



$\Sigma H = 0$

$30 + 20 - FCF \cos 45 = 0$

$FCF = 70.69KN$ (T)

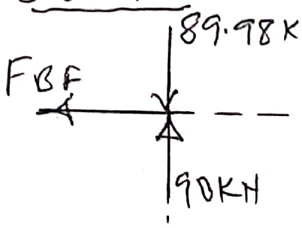
$\Sigma V = 0$

$-40 - FCB - FCF \sin 45 = 0$

$-40 - FCB - 70.69 \sin 45 = 0$

$FCB = -89.98KN$ (C)

Joint B :-



$\Sigma H = 0$

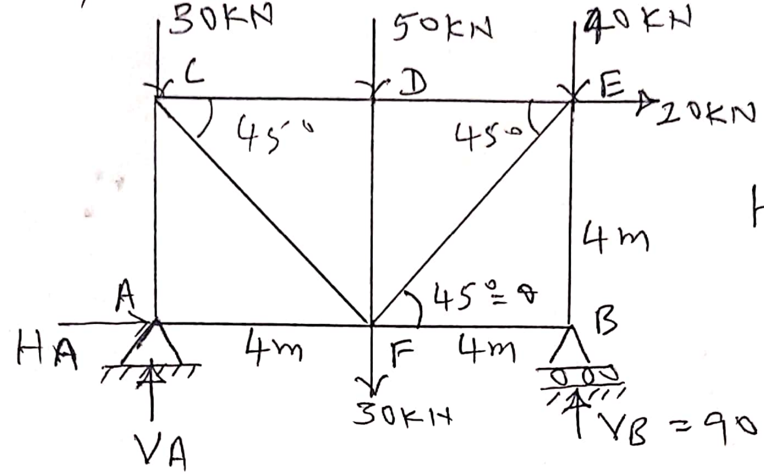
$FBF = 0$

Tabulation of Member forces.

Member	Tensile force	Compressive force
AF	20 KN	
FB	0	–
BC	–	89.98 KN
CE	–	30 KN
ED	–	30 KN
AD	–	60 KN
DF	42.42 KN	–
EF	–	50 KN
FC	70.69 KN	–

Method of Sections:-

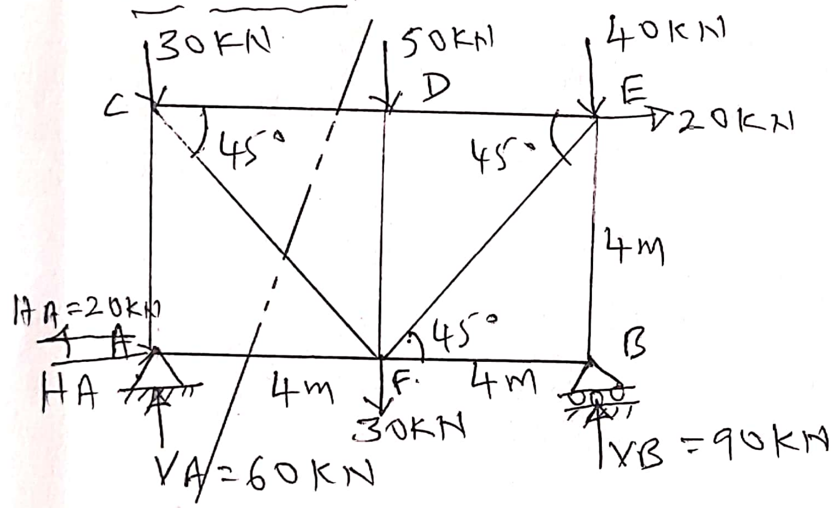
1) Determine the force in the members CD, CF and AF for the truss shown.



$\tan\theta = \frac{4}{4}$

$\theta = 45°$

Solution:-



$\Sigma M_A = 0$

$50\times4 + 40\times8 + 30\times4 - V_B\times8 + 20\times4 = 0$

$V_B = 90\,KN$

$\Sigma V = 0$

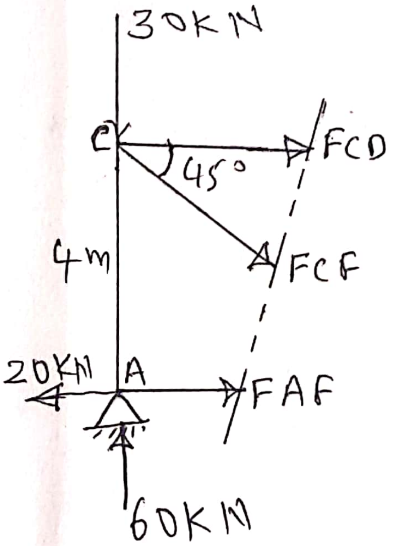
$V_A - 30 - 50 - 40 + V_B - 30 = 0$

$V_A = 60\,KN$

$\Sigma H = 0$

$H_A + 20 = 0$

$H_A = -20\,KN\ (\leftarrow)$



$\Sigma V = 0$

$-30 + 60 - F_{CF}\sin45 = 0$

$F_{CF} = 42.42\,KN\ (T)$

$\Sigma M_A = 0$

$F_{CD}\times4 + F_{CF}\cos45\times4 = 0$

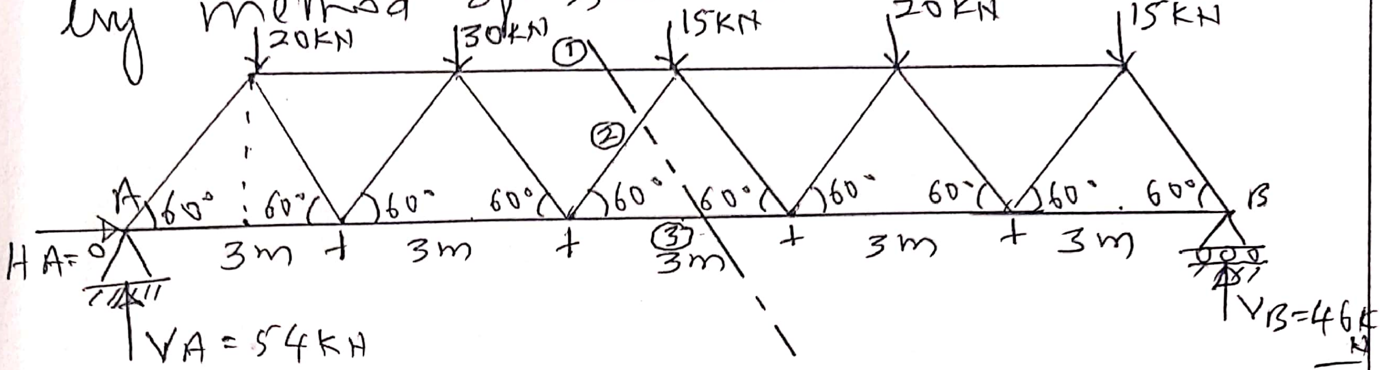
$F_{CD} = -30\,KN\ (C)$

$\Sigma H = 0$

$F_{AF} + F_{CF}\cos45 + F_{CD} - 20 = 0$

$F_{AF} = 20\,KN\ (T)$

2) Determine the forces in the members ①, ② and ③ of the truss shown in fig. by method of sections.



$\Sigma MA = 0$

$20 \times 1.5 + 30 \times 4.5 + 15 \times 7.5 + 20 \times 10.5 + 15 \times 13.5 - V_B \times 15 = 0$

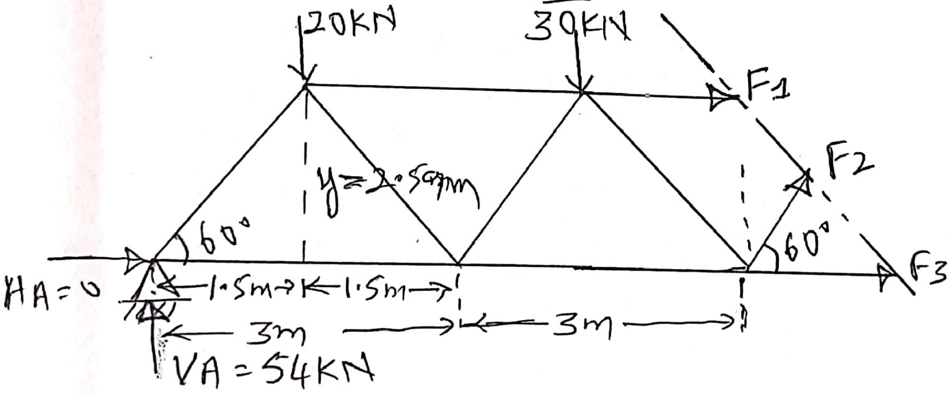
$V_B = 46 KN$

$\Sigma V = 0$

$V_A - 20 - 30 - 15 - 20 - 15 + V_B = 0$

$V_A = 54 KN$

$\Sigma H = 0, \quad H_A = 0$



$\tan 60 = \frac{y}{1.5}$

$y = 2.59m$

$\Sigma Y = 0$

$54 - 20 - 30 + F_2 \sin 60 = 0$

$F_2 = -4.61 KN \ (C)$

$\Sigma MA = 0$

$20 \times 1.5 + 30 \times 4.5 + F_1 \times 2.59 - F_2 \sin 60 \times 6 = 0$

$F_1 = -72.95 KN \ (C)$

$\Sigma H = 0$

$F_3 + F_2 \cos 60 + F_1 = 0$

$F_3 + (-4.61) \cos 60 + (-72.95) = 0$

$F_3 = 75.25 KN \ (T)$